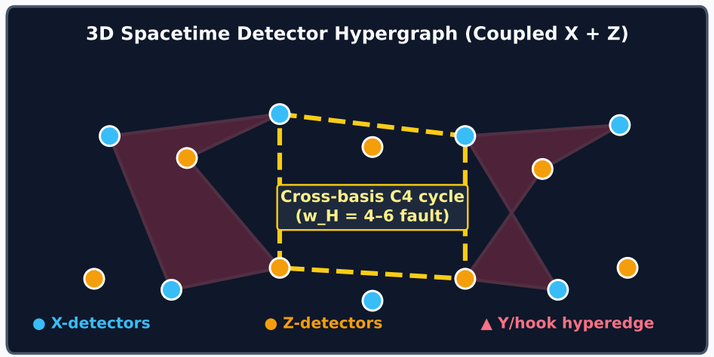
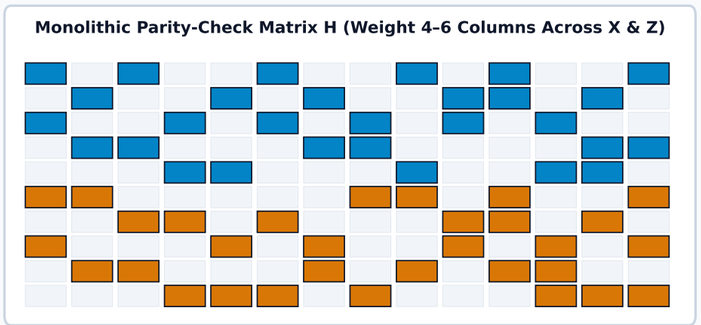
Detector Error Model (DEM) Structural Transformations: [B] Monolithic vs [G] Gari Gauge Elimination vs [M]/[O] Multi-Pass Shattering

(a) [B] Monolithic DEM (Coupled X/Z Hypergraph)

Degree 4–6 cross-basis Y/hook hyperedges & dense short 4-cycles (C4)

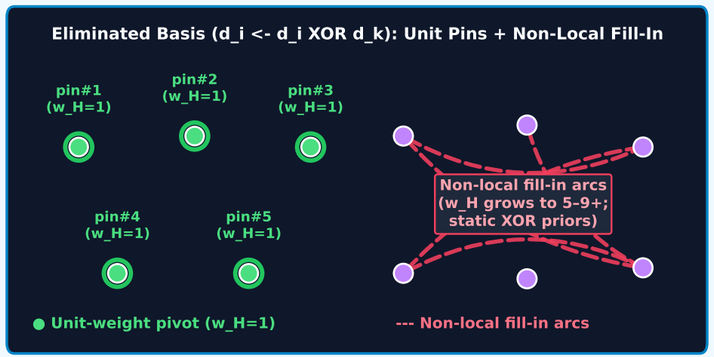


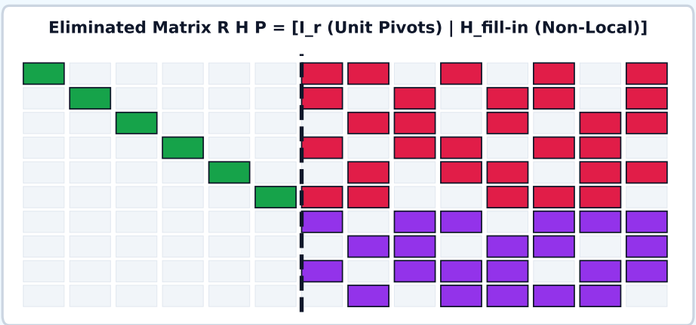


- Search complexity: Wide beam (b = 20); every fired detector branches over 4–6 coupled fault columns -> exponential A* queue growth.
- CC d=7: 42.50 ms/round, 5/8,000 errors (LER = 8.93e-05)
- BB [[72,12,6]]: 201.03 ms/round, 4/6,000 errors
- Mitten [[150,30,10]]: 5.37 s/round, 0/200 errors

(b) [G] Gari Gauge Elimination (R H P = [I_r | H_fill-in])

Offline F2 row XORs create unit-weight pivots (w_H=1) + non-local column fill-in

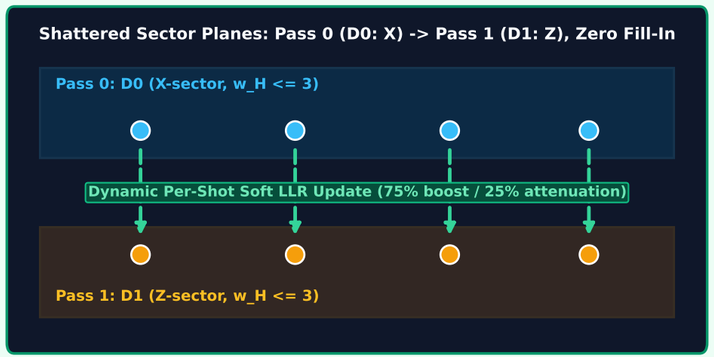


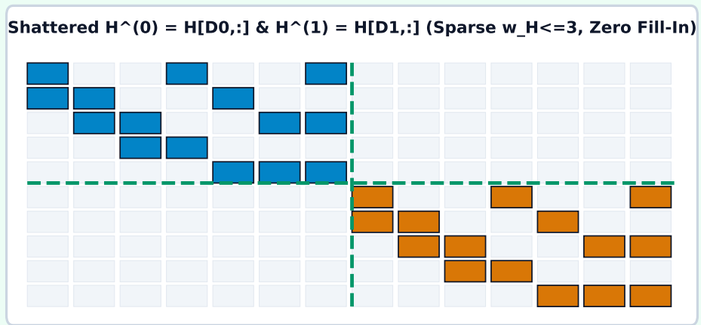


- Why [G] speeds up A* (8x–166x): Unit-weight pins (w_H=1) ordered first resolve greedily at narrow beam (b=5) with zero A* branching: CC d=5: 21.7x; CC d=7: 12.4x (3.42 ms); BB d=6: 36.4x; Mitten: 165.7x.
- Structural limitation: Row XORs create non-local fill-in (w_H -> 5–9+) & static XOR priors (CC d=5 LER 1.19x [B]; dense 4-cycles block BP).

(c) [M] / [O] Multi-Pass DEM Shattering (D0 -> D1)

Zero row elimination, zero fill-in, preserved locality + dynamic per-shot LLR update





- Zero Fill-In + Dynamic Conditioning: Preserves spatial locality & code distance while eliminating cross-basis 4-cycles (C4) for both A* and BP-OSD.
- [M] Multi-Pass A*: 357x faster than [B] on CC d=7 (119 µs, 28.7x faster than [G]); 375x faster on BB d=6 (10.3x vs [G]).
- [O] Multi-Pass BP-OSD: 610x faster on CC d=7 (70 µs); 12,819x on Mitten.